Proximity to Academic Ophthalmology and the Severity of Diabetic Retinopathy: A National County-Level Analysis of 3,119 United States Counties

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Abstract

Objective: To test whether county-level proximity to academic ophthalmology infrastructure is associated with the severity of diagnosed diabetic retinopathy (DR), and to reconcile a national population-based test with clinic-based studies reporting that greater distance from ophthalmology clinics predicts proliferative disease.

Design: Cross-sectional analysis of county-level surveillance estimates.

Subjects: 3,119 of 3,139 United States counties with 2021 surveillance estimates, representing 36.1 million adults with diabetes and 9.53 million with diagnosed DR.

Methods: Road-network drive times were computed from population-weighted county centroids to each of 123 ACGME-accredited ophthalmology residency programs accredited by 2021, and separately to the nearest county containing any patient-care ophthalmologist. The outcome was the proportion of diagnosed DR that was vision-threatening, from the CDC Vision and Eye Health

Surveillance System. Models were quasibinomial and estimated within county × race/ethnicity × age strata. Four prespecified tests addressed whether any association is attributable to academic centers specifically rather than to eye care supply in general.

Main Outcome Measures: Vision-threatening share of diagnosed DR.

Results: The vision-threatening share did not rise with distance from an academic program. It fell slightly and monotonically, from 18.9% in counties within 30 minutes of a program to 16.1% in counties beyond four hours. The adjusted association was small and negative ($\beta = -0.017$ per SD of log drive time, $p < 0.001$) and did not attenuate under adjustment for provider supply, social vulnerability, Medicare diagnostic volume, or preventive screening. Counties with local ophthalmologists but far from an academic center had a vision-threatening share of 18.6%, against 19.6% for those near one. Academic proximity did not matter more where no local ophthalmologist existed (interaction $p = 0.70$), and capacity-weighted access did not perform better than simple proximity.

Conclusions: Proximity to academic ophthalmology does not predict less severe diabetic retinopathy at the population level. The pattern is more consistent with diagnostic intensity, where subspecialty capacity concentrates, than with clinical benefit. Clinic-based findings that distance predicts proliferative disease are likely to reflect referral selection, which a population-based design does not reproduce. Geographic placement of residency programs is not supported by these data as a lever for reducing vision-threatening diabetic eye disease.

1 Introduction

Academic ophthalmology departments concentrate subspecialty retinal capacity. Panretinal photocoagulation, intravitreal anti-VEGF therapy, and vitrectomy for tractional detachment are delivered disproportionately at teaching institutions, and it is intuitive that communities distant from that capacity should carry more advanced diabetic eye disease. This intuition underlies proposals to site new training programs as a means of addressing geographic disparities in vision loss.

Recent evidence appears to support it. Xie and colleagues, studying 73,618 patients with diabetes at two academic centers, found that patients living more than eight miles from the ophthalmology clinic they attended had substantially higher odds of proliferative diabetic retinopathy than patients living within eight miles of the same clinic, across every quartile of neighborhood deprivation.[13] That is a large, carefully conducted study, and its distance finding is striking.

There is, however, a structural reason to be cautious about generalizing it. The exposure is distance to the clinic *at which the patient was seen*, and the sample consists of patients who attended a tertiary referral center. Patients living far from such a center who present there are, by construction, unusual: mild disease is typically managed locally, and it is the severe cases that travel. The authors say so themselves, noting that their finding may reflect referral bias, with local providers managing mild cases while proliferative disease is referred inward.[13] Under that mechanism, distance would predict severity among attendees without distance causing severity in the population.

Distinguishing the two requires a design that does not condition on attendance at an academic center. We therefore test the same hypothesis on the full population of United States counties, where every county is observed regardless of whether its residents travel to a teaching hospital, and where the outcome is measured by surveillance rather than by clinic encounter.

We ask three questions. First, does the severity of diagnosed diabetic retinopathy rise with distance from an academic ophthalmology program? Second, if any association exists, is it attributable to academic centers specifically, or merely to the fact that academic centers are located where ophthalmologists are? Third, does the observed pattern behave as clinical benefit would, or as differential detection would?

2 Methods

This study follows the STROBE reporting guideline for cross-sectional studies. All data were publicly available, aggregated, and contained no individual-level identifiers; institutional review board approval was not required. The full data pipeline is publicly available, and all sources are described in detail in the accompanying Supplementary Appendix.

2.1 Units of analysis and eligibility

The unit of analysis was the county or county-equivalent jurisdiction. VEHSS publishes 2021 diabetic retinopathy estimates with both all-stage and vision-threatening case counts for 3,139 counties; Puerto Rico is absent from the source entirely. Twenty of these were excluded, leaving 3,119 in the analytic sample: twelve had no routable road path to any residency program (eight Alaska boroughs and four Hawaii counties); four carried retired or renamed FIPS codes present in the surveillance file but absent from 2021 Census geography (Valdez-Cordova and Wade Hampton in Alaska, Shannon County in South Dakota, and Bedford city in Virginia); and four had insufficient age-stratified case counts to enter the stratified models (Petroleum County, Montana, and Arthur, Blaine and McPherson Counties, Nebraska).

2.2 Outcome

The outcome is the *vision-threatening share*: the number of people with vision-threatening DR divided by the number with DR of any stage, among adults with diabetes, from the CDC Vision and Eye Health Surveillance System (VEHSS) composite prevalence estimates for 2021.[3, 9] VEHSS defines vision-threatening DR as severe non-proliferative DR, proliferative DR, or diabetic macular edema.

We use the severity share rather than DR prevalence deliberately. Prevalence among people with diabetes is dominated by glycemic exposure and disease duration. The share of existing disease that has reached a vision-threatening stage is a case-mix measure that speaks to whether disease was detected and treated before progression, which is the quantity the access hypothesis concerns. Aggregating our county numerators reproduces the published national figures (DR 26.43% of adults with diabetes against a published 26.43%; VTDR 5.05% against 5.06%).

Because VEHSS constructs county estimates by applying national age-, sex-, and race-specific rates to each county's demographic composition, county demographics move the published estimate mechanically.[9] All models are therefore estimated within county × race/ethnicity × age strata with stratum fixed effects (non-Hispanic White, non-Hispanic Black, and Hispanic adults, each aged 40–64 and 65–84; 13,035 cells across 3,119 counties), so that composition cannot contribute to any estimated association.

2.3 Exposures

Academic proximity. Locations of ophthalmology residency programs were compiled from FRIEDA/ACGME listings, geocoded, and assigned to counties by point-in-polygon overlay against 2021 TIGER/Line boundaries. Of 128 programs, 123 accredited in or before 2021, located in 90 counties, were retained for this 2021 cross-section. Drive times were computed over the road network using the Open Source Routing Machine from each county's population-weighted centroid, taken from the Census Bureau's 2020 Centers of Population file.[11] Counties containing a program were assigned zero travel burden.

General eye care proximity. Because academic programs are sited where ophthalmologists practice, distance to an academic program conflates two things. We therefore computed a second exposure: drive time to the nearest county containing at least one patient-care ophthalmologist, using 2021 counts from the Area Health Resources File.[8] Of the analytic counties, 1,950 (62.5%) contained no patient-care ophthalmologist at all.

2.4 Covariates

Ophthalmologist and optometrist density per 100,000 residents (AHRF 2021), population density from TIGER/Line land area and 2021 American Community Survey population, the 2013 Rural-Urban Continuum Code, and the overall percentile ranking of the 2020 CDC/ATSDR Social Vulnerability Index.[4] For the detection analyses we additionally used county Medicare imaging, evaluation and management, and diagnostic test events per 1,000 beneficiaries from the CMS Medicare Geographic Variation file,[5] and county mammography screening completion from County Health Rankings.[6]

2.5 Statistical analysis

Models were quasibinomial generalized linear models with vision-threatening cases as successes and all-stage DR cases as trials; county aggregates of this kind are overdispersed, and the quasibinomial family scales standard errors accordingly. Continuous predictors were standardized and drive times log-transformed, so coefficients are per standard deviation and comparable across exposures.

Four prespecified tests addressed the specificity question: (i) the academic coefficient conditional on general eye care proximity; (ii) a 2×2 comparison of counties with and without a local ophthalmologist crossed with academic proximity; (iii) an interaction testing whether academic proximity matters more where no local ophthalmologist exists; (iv) a capacity-weighted gravity access score using ophthalmology resident counts with a 120-minute Gaussian decay, replacing simple proximity; All four were estimated with the full covariate set (social vulnerability, ophthalmologist and optometrist density, population density, rurality, distance to the nearest ophthalmologist, and stratum fixed effects); unadjusted estimates are reported alongside where they differ materially.

Note on orientation: drive time and the gravity access score run in opposite directions, since longer drive time means worse access while a higher access score means better access. A negative drive-time coefficient and a positive access-score coefficient therefore carry the same substantive meaning. Analyses used R 4.6.0.

3 Results

3.1 Severity does not rise with distance

Median drive time to the nearest residency program was 123 minutes (interquartile range 77 to 192). The vision-threatening share did not increase with distance. It declined slightly and monotonically, from 18.9% among counties within 30 minutes of a program to 16.1% among counties more than four hours away (Figure 1). The access hypothesis predicts the opposite slope.

3.2 The association is small, negative, and does not attenuate

In the within-stratum model the coefficient on log drive time was -0.017 per standard deviation ($p < 0.001$). It was essentially unchanged by adjustment for general eye care proximity, provider

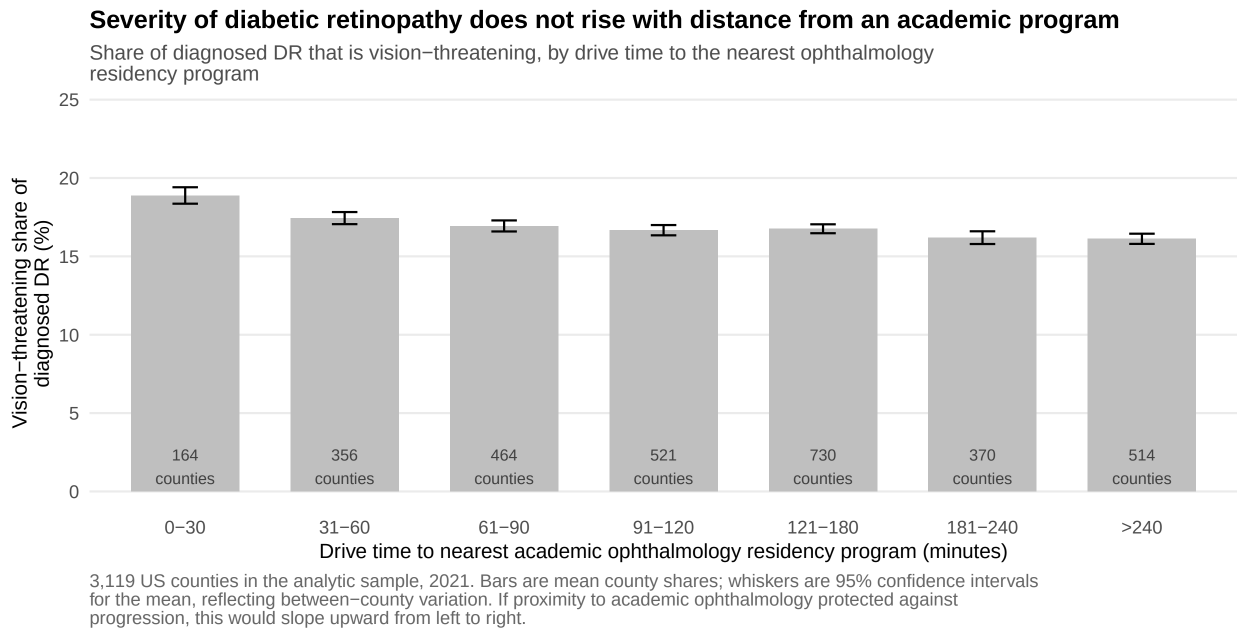

Figure 1: Vision-threatening share of diagnosed diabetic retinopathy by drive time to the nearest academic ophthalmology residency program.

density, population density, rurality, social vulnerability, Medicare diagnostic volume, and preventive screening completion (Figure 2). A confounded association would be expected to move toward zero under such adjustment; this one does not, and it points the wrong way throughout.

Entered jointly, the two exposures behaved in opposite directions. Distance to *any* ophthalmologist carried the positive sign an access hypothesis predicts ($\beta = +0.014$, $p = 0.06$), while distance to an *academic* program carried a negative one ($\beta = -0.023$, $p < 0.001$). The two exposures correlate at only 0.41 and are therefore separately identified.

3.3 Discordant counties

Counties with local ophthalmologists but far from an academic center, the group whose disadvantage the access hypothesis most directly predicts, did not have worse case mix: 18.6% vision-threatening, against 19.6% among counties with local ophthalmologists near an academic center (Figure 3). The larger contrast in the table is horizontal rather than vertical: counties with any local ophthalmologist had roughly two percentage points *more* vision-threatening disease than counties with none.

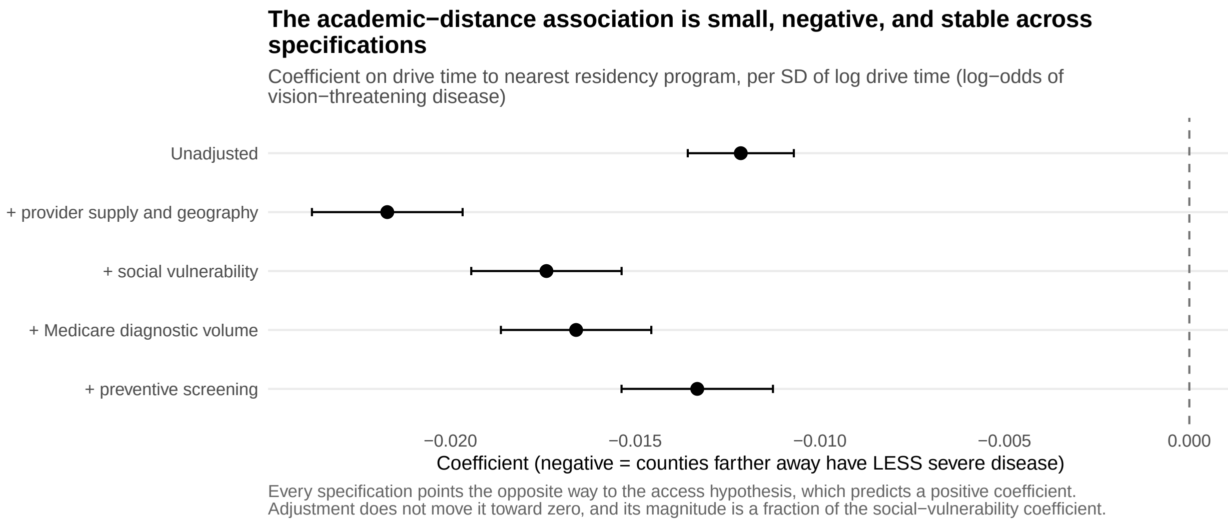

Figure 2: Coefficient on drive time to the nearest residency program across progressively adjusted specifications.

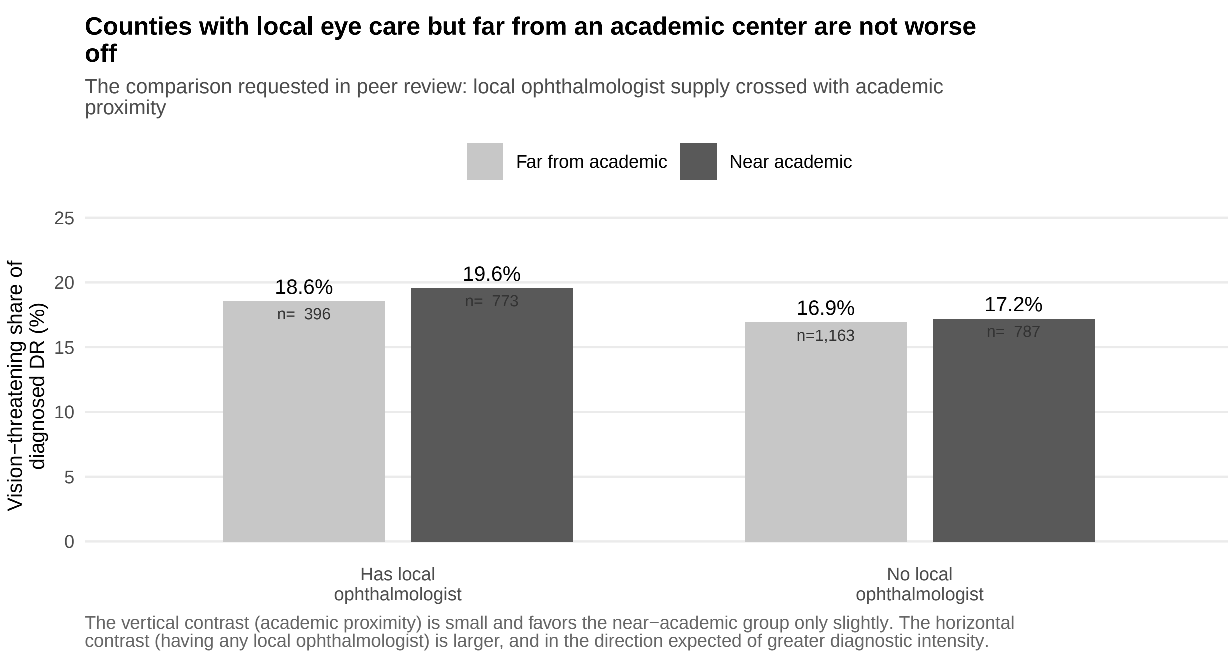

Figure 3: Vision-threatening share by local ophthalmologist supply and academic proximity.

3.4 Academic proximity does not matter more where there is no alternative

If academic centers supply something distinctive, their reach should matter most where no local eye care exists. The interaction between academic drive time and having no local ophthalmologist was null ($\beta = 0.008$, $p = 0.70$).

Replacing simple proximity with a capacity-weighted gravity score did not help. Because a higher access score denotes *better* access, its positive coefficient ($\beta = +0.018$, $p < 0.001$) carries the same substantive meaning as the negative drive-time coefficient: counties with better modeled access to academic capacity had a higher, not lower, vision-threatening share. We emphasize that this is not independent confirmation. The two exposures are alternative encodings of the same underlying geography and correlate at 0.61 once orientation is aligned; entered jointly both attenuate (drive time -0.016, access score $+0.009$).

3.5 Spatial confirmation

Residual spatial dependence was substantial (Moran's $I = 0.237$, $p < 0.001$). Under a Leroux conditional autoregressive model fitted to the largest stratum, the academic drive time posterior mean was -0.010 (95% credible interval -0.019 to -0.001), still negative and still small.

4 Discussion

Across every specification we examined, proximity to academic ophthalmology failed to predict less severe diabetic retinopathy. The association ran in the opposite direction to the access hypothesis, was small, did not attenuate under adjustment, was absent where it should have been strongest, and was not improved by weighting for program capacity.

Several results point toward detection rather than clinical benefit. Counties containing any ophthalmologist had about two percentage points more vision-threatening disease than counties with none. Counties with higher Medicare imaging volume had less vision-threatening disease, consistent with greater diagnostic activity identifying more mild cases and enlarging the denominator. The authors of the underlying surveillance estimates anticipated this, cautioning that claims diagnoses are influenced by access to ophthalmologists and that their model may mistake such factors for true variation in prevalence.[9]

4.1 Reconciling with clinic-based findings

Our result appears to contradict Xie et al., who found that patients living more than eight miles from their ophthalmology clinic had markedly higher odds of proliferative disease.[13] We do not think the two are in conflict; they estimate different quantities.

Their exposure is distance to the clinic the patient attended, in a sample of patients who attended a tertiary referral center. Selection into that sample depends jointly on distance and on severity: a patient living an hour away is far more likely to appear in a referral center's records if their disease is proliferative than if it is mild, because mild disease is managed locally. The association between distance and severity is therefore partly generated by who enters the sample. The authors identify this possibility explicitly.

Our design does not condition on attendance. Every county is observed whether or not its residents travel to a teaching hospital, and the outcome is drawn from surveillance covering the whole diabetic population rather than from clinic encounters. Under that design the distance association disappears and reverses. The natural reading is that the clinic-based association reflects referral selection rather than a population-level effect of travel burden, and that inferences about workforce geography should not be drawn from referral-center samples without a population-level check.

This distinction matters practically. If distance from academic centers caused proliferative disease, siting new programs would be a rational intervention. If the clinic-based association is a selection artifact, program placement would be expected to change where severe disease is *seen* without changing how much of it there is.

4.2 What the data do support

Two findings survive. First, the presence of any local ophthalmologist tracks measured disease severity in the direction expected of diagnostic intensity, and does so more consistently than the academic character of the nearest institution. Second, in a companion analysis of the same data, county social vulnerability predicts the vision-threatening share within demographic strata and is substantially attenuated by general preventive-screening completion, implicating screening engagement rather than specialty geography.

4.3 Limitations

First, this is an ecological analysis; associations cannot be interpreted at the individual level. Second, the outcome is derived from modeled estimates built on diagnosed disease in claims, so undiagnosed disease is invisible; this is precisely the mechanism we invoke to explain the pattern, and we cannot fully separate it from a true absence of effect. Third, drive times model private vehicle travel and ignore traffic, ferries, and public transportation entirely; twelve island and remote counties in Alaska and Hawaii had no routable path and are excluded, along with four counties carrying retired FIPS codes and four with insufficient age-stratified counts. Fourth, residency program location is a proxy for academic ophthalmology and does not capture non-academic retina practices, satellite clinics, or teleretinal programs. Fifth, Puerto Rico is absent from the VEHSS source.

4.4 Conclusion

At the population level, counties farther from academic ophthalmology do not carry more vision-threatening diabetic retinopathy. The geography of academic ophthalmology appears to shape where advanced disease is detected rather than how much of it occurs. Strategies aimed at reducing preventable vision loss are better directed at screening completion within disadvantaged communities than at the geographic placement of training programs.

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